

backend/core/profiler.py Documentation (Roman Urdu)

Yeh file application ka chota sa "Hardware Checker" hai.

File Ka Maqsad (Purpose)

Iska principle maqsad user ka PC (Windows/Linux) check karni hai. System main konsa Processor/CPU laga hai? GPU aur Graphics card kitnay limit ka ha VRAM kitni ha? Aur RAM (memory) kitni available hai.

Code Blocks aur Functions ki Tafseel

1. Hardware Detection (`_get_gpu_info`, `_get_cpu_info`)

Kya karta hai?:

- CPU ke liye yeh `psutil` aur Windows ki `wmic` command chalata hai ta-k naam (e.g., Intel i7) aur cores (e.g., 8 Cores) dhoond sakay.
- GPU k liye yeh standard Python library `GPUtil` use krti ha lkn mostly cases me system command `nvidia-smi` ko shell mein run krkay Graphic Card ka naam aur VRAM (e.g. 12000MB) bytes parse or clear cut get krleti ha.

2. `def _recommend_models(vram_mb, ram_mb)`

Kya karta hai?: Pura scan function (profiler) idher apny results fetch karta ha jisse logic model recommended models ka faisla krti hai.

- Agar VRAM 8GB+ hai: Yeh baray aur smart models (Llama 3.1 8B, Mistral 7B) refer karta ha.
- Agar GPU choota (4GB) ha lkn RAM bari ha: Ye Qwen/Gemma jaisy models recommend karta hai.
- Agar GPU nai ha, bus ram hi limited ha (8GB): Tou TinyLlama ya phi3.